Chuanhan Li

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Education

University of California, Santa Cruz (UCSC) *CA, USA*
PhD student @ UCSC CSE **Sep. 2024 - Now**

Huazhong University of Science and Technology (HUST) *Wuhan, China*
Master of Computer Application Technology **Sep. 2021 – Jul. 2024**

Central South University *Changsha, China*
Bachelor of Computer Science **Sep. 2017 – Jun. 2021**

Research Summary

- A second-year PhD student in computer engineering, systems and hardware security, focusing on extending confidential computing to disaggregated memory and interconnects (CXL).
- First author of a MICRO 2025 paper on efficient security support for CXL memory, built prototype mechanisms on Gem5 and evaluated overheads.
- Strong implementation background: OpenJDK runtime modifications, RDMA optimization, Gem5 simulation, performance analysis on Linux, proficiency in C/C++.

Publications and Drafts

[MICRO25] **Chuanhan Li**, Jishen Zhao, Yuanchao Xu, "Efficient Security Support for CXL Memory through Adaptive Incremental Offloaded (Re-)Encryption", the 58th IEEE/ACM International Symposium on Microarchitecture, 2025.

[Under Review (ISCA26)] **Chuanhan Li**, Yuanchao Xu, DTS: Efficiently Securing CXL Memory via Disaggregated Tiered Security

[Under Review (ISCA26)] **Chuanhan Li**, Yuanchao Xu, Scalable CXL Link-level Integrity and Data Encryption for Multi-host

Research Projects

Data Serialization Optimization in RDMA Network **Feb. 2022 - Jun. 2022**

- Designed a serialization scheme that preserves in-memory object layout, eliminating pre-transmission copies.
- Cut serialization latency by 42% on microbenchmarks.

ZipGC: Compressed Heap in Managed Runtime atop Disaggregated memory **Feb. 2023 - Feb. 2024**

- Extended OpenJDK's ZGC to identify cold objects and apply base-delta compression.
- Achieved 2.5× bandwidth reduction with <7% runtime overhead on SPECjbb.

AIORE: Efficient Security Support for CXL Memory through Adaptive Incremental Offloaded (Re-)Encryption **Sep. 2024 - Jun. 2025**

- Designed adaptive encryption modes switching between AES-XTS and AES-CTR on CXL secure memory.

- Detailed analysis on the pros and cons of AES-XTS and AES-CTR under different memory access behavior characteristics.

DTS: Efficiently Securing CXL Memory via Disaggregated Tiered Security **Mar. 2025 - Now**

- Designed adaptive memory encryption with page placement awareness.

Scalable CXL Link-level Integrity and Data Encryption for Multi-host CXL Shared Disaggregated Memory **Sep. 2025 - Now**

- Designed efficient secure CXL Link Data Encryption and Integrity support through cryptographic pad reuse and merge of AES-GCM.

Research Experience

Graduate Researcher - China Grid Computing Laboratory, HUST **Feb. 2021 - Jul. 2024**

- Led two performance improvement studies on a disaggregated memory system with RDMA stack and OpenJDK.
- **Advisor:** Prof. Haikun Liu, Prof. Chencheng Ye

Research Assistant - Center for Research in Systems and Storage, UCSC **Sep. 2024 - Now**

- First author of a MICRO 2025 paper, building secure CXL memory mechanisms and evaluating on representative workloads.
- **Advisor:** Prof. Yuanchao Xu

Skills

Programming Languages: C, C++, Java, Python, Shell
Familiarity: Confidential Computing (AMD SEV/Intel TDX & SGX), AES Encryption Algorithm, RDMA, Linux scripting